SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Preprocessing techniques like Data cleaning, Data integration,Data transformation and data Reduction, for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure A
SENARIO BASED QUESTIONS

Criteria	Understanding of Scenario ***(2)***	Identification of Key Issues ***(2)***	Application of Concepts ***(2.5)***	Decision Making ***(2)***	Clarity of Response ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **J.Mohamed Aseem(192511258)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

J.Mohamed Aseem
Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SCENARIO BASED QUESTIONS

Q1. Data Preprocessing in Retail Analytics (*L3, L4)*

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

Customer_ID	Age	Gender	Monthly_Spend (₹K)	Purchase_Frequency	Loyalty_Level
C1	23	M	20	5	Low
C2	NaN	Male	35	8	Medium
C3	31	F	NaN	10	High
C4	105	Female	500	50	High
C5	27	male	25	NaN	Low
C6	40	FEMALE	45	12	Medium
C7	36	M	38	9	Low
C8	42	Female	300	45	High

Tasks

A. Handling Missing Values
i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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B. Outlier Detection and Treatment
i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization
i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioural attributes, leading to redundancy and increased computational cost.

Sample Dataset

Cust_ ID	Age	Income (₹K)	Purchase _Value (₹K)	Credit_ Score	Orde rs	Avg_Bal ance (₹K)	EMI (₹)	Customer _Class
U1	24	40	100	650	15	12	5000	Low
U2	34	60	150	700	20	18	7000	Medium
U3	44	85	210	720	28	25	9000	Medium
U4	52	120	320	800	35	40	15000	High
U5	29	45	110	680	18	14	5500	Low
U6	39	75	180	710	24	22	8500	Medium
U7	56	150	360	820	40	50	18000	High
U8	37	70	160	705	22	20	8000	Medium

Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)

ii. Interpret whether the relationship is **positive or negative**

iii. Explain how covariance helps in identifying feature relationships

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B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**

ii. Select an optimal subset of features for modeling

iii. Justify your selection

C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)

ii. Explain why normalization/standardization is required

Q3. Data Transformation & Discretization in Education Analytics *(L3, L4, L5)*

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

Student_ID	Study_Hours	Test_Score	Attendance (%)	Performance
S1	2	40	60	Poor
S2	3	50	65	Average
S3	4	55	70	Average
S4	5	65	75	Good
S5	6	75	80	Good
S6	7	85	85	Good
S7	8	90	88	Excellent
S8	9	95	90	Excellent
S9	10	98	92	Excellent
S10	11	100	95	Excellent

Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score

ii. Calculate bin width and define intervals

B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**

ii. Assign values accordingly

C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

4. Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

5. Data Transformation in Online Shopping Behavior Analysis *(L3, L4, L5)*
An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.
To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

Sample Dataset: Visit Duration

Customer_ID	Visit_Duration (seconds)
C1	15
C2	30
C3	45
C4	60

C5	120
C6	300
C7	600
C8	900
C9	1800
C10	3600

Tasks

A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records

ii. Assign each value to its respective bin

iii. Analyse how effectively this method groups short and long visit durations

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B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration

ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly

iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

Answers

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

i. Identification of missing values

The missing values in the given dataset are:

Customer ID	Attribute	Missing value
C2	Age	NaN
C3	Monthly_Spend	NaN
C5	Purchase_Frequency	NaN

ii. Applying imputation techniques

Method 1: Median imputation:

Median imputation replaces missing numerical values with the median of the available values.

- Age values: 23, 31, 15, 27, 40, 36, 42. Sorted: 15, 23, 27, 31, 36, 40, 42.
- Median Age = 31 years.
- Monthly Spend values: 20, 35, 500, 25, 45, 38, 300. Sorted: 20, 25, 35, 38, 45, 300, 500.
- Median Monthly Spend = ₹38K.
- Purchase Frequency values: 5, 8, 10, 50, 12, 9, 45. Sorted: 5, 8, 9, 10, 12, 45, 50.
- Median Purchase Frequency = 10.

Method 2: k-NN imputation:

k-Nearest Neighbors (k-NN) imputation estimates missing values using similar records based on other available attributes. For example, the missing Age for C2 can be estimated from customers with similar spending, purchase frequency, and loyalty levels.

The exact k-NN estimates depend on the distance metric, feature scaling, and selected value of \(k\). Therefore, median imputation is used for the numerical results below.

iii. Imputed values and justification

Customer ID	Attribute	Imputed value
C2	Age	31
C3	Monthly_Spend	₹38K
C5	Purchase_Frequency	10

Median imputation is appropriate because it is simple and less affected by extreme values. k-NN imputation may preserve relationships between attributes, but it requires more computation and careful selection of similar records.

B. Outlier Detection and Treatment

i. Detecting outliers using the Z-score method

The Z-score formula is:

\[Z=\frac{X-\mu}{\sigma} \]

Where \(X\) is the observed value, \(\mu\) is the mean, and \(\sigma\) is the standard deviation. A common rule identifies values with \(|Z|>3\) as potential outliers.

ii. Identifying extreme values

- Age = 105 is an extreme and likely invalid value in the sample.
- Monthly Spend = ₹500K and ₹300K are much higher than most other spending values.

- Purchase Frequency = 50 and 45 are also unusually high compared with most records.

The Z-score method should be applied to the numerical columns, including the imputed values, and the calculated scores should be checked before confirming outliers. The age value of 105 mentioned in the task is not present in the displayed sample; the sample shows 15 for C4.

iii. Outlier treatment

Attribute	Treatment	Reason
Age	Validate and correct or remove invalid records	Prevents unrealistic customer ages
Monthly Spend	Apply capping or log transformation	Reduces the influence of extreme spending
Purchase Frequency	Validate and cap if necessary	Reduces the effect of unusually high counts

Capping limits extreme values to a chosen threshold, while log transformation reduces right-skewness. Records should be removed only when the values are confirmed to be invalid rather than merely unusual.

C. Categorical Data Standardization:

i. Identifying inconsistencies

The Gender column contains inconsistent formats such as M, Male, male, F, Female, and FEMALE.

ii. Standardized format

Original value	Standardized value
M	Male
Male	Male
male	Male
F	Female
Female	Female
FEMALE	Female

Standardization ensures that the same category is not treated as several different values during analysis.

Conclusion: Data preprocessing improves data quality by handling missing values, reducing outlier effects, and standardizing categories. The cleaned dataset will support more reliable customer recommendations.

Q2. Covariance and Data Reduction in E-commerce Analytics:

A. Covariance Analysis:

i. Covariance between Income and Purchase_Value

The formula for sample covariance is:

$$\operatorname{Cov}(X,Y)= \frac{\sum (X_i-\bar X)(Y_i-\bar Y)}{n-1}$$

Given Income values: 40, 60, 85, 120, 45, 75, 150, 70.

Purchase_Value values: 100, 150, 210, 320, 110, 180, 360, 160.

Number of records: $n=8$.

Mean Income:

$$\bar X=\frac{645}{8}=80.625$$

Mean Purchase Value:

$$\bar Y=\frac{1590}{8}=198.75$$

The sum of the cross-products of deviations is:

$$\sum (X_i-\bar X)(Y_i-\bar Y)=23743.75$$

Therefore,

$$\operatorname{Cov}(X,Y)= \frac{23743.75}{7} =\boxed{3391.96}$$

ii. Interpretation

The covariance is positive (approximately 3391.96), indicating that Income and Purchase_Value tend to increase together. Customers with higher income generally show higher purchase values in this sample.

iii. Importance of covariance

- Identifies whether two numerical attributes vary together.
- Helps discover relationships between financial and behavioural features.
- Supports feature selection and redundancy analysis.
- Helps identify attributes that may provide overlapping information.

Covariance depends on the units and scales of the variables, so correlation is often more suitable for comparing the strength of relationships.

B. Feature Selection:

i. Identifying redundant attributes

Potentially related attributes include:

- Income and Purchase_Value.
- Income and Average_Balance.
- Orders and Purchase_Value.
- Income and EMI.

These relationships should be checked using correlation analysis before removing features.

ii. Selected subset of features

Feature	Decision	Reason
Income	Retain	Important financial indicator
Purchase_Value	Retain	Directly represents spending
Credit_Score	Retain	Indicates creditworthiness
Orders	Retain	Represents purchase activity
Average_Balance	Review	May overlap with Income
EMI	Review	May overlap with other financial features
Age	Retain	Provides demographic information

Cust_ID should be excluded from predictive features because it is an identifier. Customer_Class should be retained as the target variable.

iii. Justification

Feature selection reduces unnecessary dimensions, improves computational efficiency, and may reduce overfitting. Attributes should be removed only after checking their predictive usefulness and redundancy.

C. Data Transformation:

i. Standardization

Standardization converts each numerical feature into a Z-score:

$$ Z=\frac{X-\mu}{\sigma} $$

For example, the Income mean is 80.625. Using the population standard deviation of approximately 34.88:

For U1, Income = 40.

$$ Z=\frac{40-80.625}{34.88}\approx -1.16 $$

For U7, Income = 150.

$$ Z=\frac{150-80.625}{34.88}\approx 1.99 $$

The same formula is applied separately to Age, Income, Purchase_Value, Credit_Score, Orders, Average_Balance, and EMI. Cust_ID and Customer_Class are excluded.

ii. Importance of standardization

- Brings numerical features onto a comparable scale.
- Prevents large-scale variables from dominating distance calculations.
- Improves the performance of distance-based algorithms such as k-NN and clustering.
- Supports more effective comparison of customer attributes.

Conclusion: Covariance analysis identifies relationships between features, while feature selection and standardization reduce redundancy and improve predictive modelling efficiency.

Q3. Data Transformation and Discretization in Education Analytics:

A. Equal-Width Binning:

i. Applying three bins

Equal-width binning divides the range of values into intervals of equal size.

The formula is:

$$\text{Bin Width}=\frac{\text{Maximum}-\text{Minimum}}{\text{Number of Bins}}$$

ii. Study_Hours

Minimum = 2, Maximum = 11, Number of bins = 3.

$$\text{Bin Width}=\frac{11-2}{3}=3$$

Bin	Interval	Student IDs
Bin 1 – Low	2 to less than 5	S1, S2, S3
Bin 2 – Medium	5 to less than 8	S4, S5, S6
Bin 3 – High	8 to 11	S7, S8, S9, S10

Test_Score

Minimum = 40, Maximum = 100.

$$\text{Bin Width}=\frac{100-40}{3}=20$$

Bin	Interval	Student IDs
Bin 1	40 to less than 60	S1, S2, S3
Bin 2	60 to less than 80	S4, S5
Bin 3	80 to 100	S6, S7, S8, S9, S10

B. Equal-Frequency Binning

i. Dividing the data into three bins

Equal-frequency binning divides sorted values into groups containing approximately equal numbers of records.

ii. Study_Hours

Bin	Values	Student IDs
Bin 1 – Low	2, 3, 4	S1, S2, S3
Bin 2 – Medium	5, 6, 7	S4, S5, S6
Bin 3 – High	8, 9, 10, 11	S7, S8, S9, S10

Test_Score

Bin	Values	Student IDs
Bin 1	40, 50, 55	S1, S2, S3
Bin 2	65, 75, 85	S4, S5, S6
Bin 3	90, 95, 98, 100	S7, S8, S9, S10

The ten records are distributed as 3, 3, and 4 because ten cannot be divided equally into three whole-record groups.

C. Concept Hierarchy Construction:

i. Study_Hours hierarchy

Study Hours	Category
2–4	Low
5–7	Medium
8–11	High

ii. Test_Score hierarchy

Test Score	Category
40–59	Poor
60–79	Average
80–100	Excellent

Concept hierarchies convert numerical values into meaningful categories. This simplifies interpretation and supports decision-making in education analytics.

Conclusion: Equal-width binning creates intervals of equal size, while equal-frequency binning creates groups with similar record counts. Both methods simplify continuous data and improve interpretability.

Q4. Data Normalization:

Given data:

$$ 200,\ 300,\ 400,\ 600,\ 1000 $$

A. Min-Max Normalization

The formula for normalizing data to the range 0 to 1 is:

$$ X'=\frac{X-X_{\min}}{X_{\max}-X_{\min}} $$

Here, $X_{\min}=200$ and $X_{\max}=1000$.

$$ X'=\frac{X-200}{1000-200} =\frac{X-200}{800} $$

Calculation:

Original value	Calculation	Normalized value
200	(200 − 200) / 800	0.000
300	(300 − 200) / 800	0.125
400	(400 − 200) / 800	0.250
600	(600 − 200) / 800	0.500
1000	(1000 − 200) / 800	1.000

Result: The normalized values are 0, 0.125, 0.250, 0.500, and 1.

B. Z-Score Normalization:

The formula is:

$$ Z=\frac{X-\mu}{\sigma} $$

Step 1: Calculate the mean:

$$\mu=\frac{200+300+400+600+1000}{5} =\frac{2500}{5}=500$$

Step 2: Calculate the population standard deviation:

$$\sigma=\sqrt{\frac{\sum(X_i-\mu)^2}{N}}$$

$$\sigma=\sqrt{\frac{90000+40000+10000+10000+250000}{5}}$$

$$\sigma=\sqrt{80000}\approx282.843$$

Step 3: Calculate the Z-scores

Original value	Calculation	Z-score
200	(200 − 500) / 282.843	−1.061
300	(300 − 500) / 282.843	−0.707
400	(400 − 500) / 282.843	−0.354
600	(600 − 500) / 282.843	0.354
1000	(1000 − 500) / 282.843	1.768

Result: The Z-score values are approximately −1.061, −0.707, −0.354, 0.354, and 1.768.

Conclusion: Min-Max normalization scales values to a fixed range of 0 to 1. Z-score normalization expresses values relative to the mean and standard deviation, producing data with a mean of approximately zero and a population standard deviation of one.

Q5. Data Transformation in Online Shopping Behavior Analysis:

A. Equal-Frequency Binning:

i. Dividing the dataset into three bins

Equal-frequency binning groups the sorted visit durations into approximately equal numbers of records.

There are 10 records, so the data is divided into groups of 3, 3, and 4 records.

ii. Assigning values to bins

Bin	Visit Duration (seconds)	Category
Bin 1	15, 30, 45	Short visits
Bin 2	60, 120, 300	Medium visits
Bin 3	600, 900, 1800, 3600	Long visits

iii. Analysis

- Bin 1 contains customers who spend very little time on product pages.
- Bin 2 contains customers who spend a moderate amount of time browsing products.
- Bin 3 contains customers who spend longer periods exploring products.
- Equal-frequency binning balances the number of records in each group, but the ranges are not equal in size.

This method simplifies customer segmentation and helps identify different browsing behaviours.

B. Z-Score Normalization

i. Calculating the mean and standard deviation

The visit durations are:

15, 30, 45, 60, 120, 300, 600, 900, 1800, 3600.

Mean:

$$\mu=\frac{\sum X}{N}=\frac{7470}{10}=747$$

Population standard deviation:

$$\sigma=\sqrt{\frac{\sum(X_i-\mu)^2}{N}}$$
$$\sigma\approx1090.92$$

ii. Applying Z-score normalization

The formula is:

$$Z=\frac{X-\mu}{\sigma}=\frac{X-747}{1090.92}$$

Customer ID	Original Duration	Z-score
C1	15	−0.671
C2	30	−0.657
C3	45	−0.643
C4	60	−0.630
C5	120	−0.575
C6	300	−0.410
C7	600	−0.135
C8	900	0.14
C9	1800	0.965
C10	3600	2.615

iii. Analysis of extreme values

- Z-scores indicate how far visit durations are from the mean.
- Values below the mean have negative Z-scores.
- Values above the mean have positive Z-scores.
- The duration of 3600 seconds has a Z-score of approximately 2.615, showing that it is considerably higher than the mean.
- Z-score normalization does not remove extreme values. It changes their scale but retains their relative differences.

Therefore, Z-score normalization helps compare visit durations, although highly skewed data may require a logarithmic transformation before normalization.

C. Min-Max Normalization

i. Applying the formula

Min-Max normalization converts visit durations into a range from 0 to 1.

$$ X'=\frac{X-X_{\min}}{X_{\max}-X_{\min}} $$

Here:

$$ X_{\min}=15,\qquad X_{\max}=3600 $$
$$ X'=\frac{X-15}{3600-15} =\frac{X-15}{3585} $$

ii. Transforming all values

Customer ID	Original Duration (seconds)	Normalized Value
C1	15	0
C2	30	0.0042
C3	45	0.0084
C4	60	0.0126
C5	120	0.0293
C6	300	0.0795
C7	600	0.1632
C8	900	0.2469
C9	1800	0.4979
C10	3600	1

iii. Comparison between Z-score and Min-Max normalization:

Feature	Z-score Normalization	Min-Max Normalization
Range	No fixed range	0 to 1
Basis	Mean and standard deviation	Minimum and maximum
Extreme values	Remain in the data	Remain in the data
Outlier sensitivity	Affected by extreme values through mean and standard deviation	Highly sensitive to minimum and maximum values
Main use	Comparing values relative to the mean	Scaling features to a fixed range

Conclusion: Equal-frequency binning groups visit durations into short, medium, and long categories. Z-score normalization expresses durations relative to the mean, while Min-Max normalization scales all durations between 0 and 1. These techniques support effective customer segmentation and clustering in online shopping analytics.